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krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ pwd
/home/krishnasree08/ossp_ubuntu_topics
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ cat > topic1_history.c <<'EOF'
#include <stdio.h>
#include <string.h>

int main(void) {
    char history[10][100];
    char input[100];
    int count = 0;

    printf("Command History Demo\n");
    printf("Type commands, :prev, :next, or :quit\n");

    while (1) {
        printf("> ");

        if (!fgets(input, sizeof(input), stdin))
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, ":quit") == 0)
            break;

        if (strcmp(input, ":prev") == 0) {
            if (count > 0)
                printf("Previous: %s\n", history[count - 1]);
            else
                printf("History is empty\n");
            continue;
        }

        if (strcmp(input, ":next") == 0) {
            printf("Enter a new command to continue.\n");
            continue;
        }

        if (strlen(input) > 0) {
            if (count < 10) {
                strcpy(history[count], input);
                count++;
EOF return 0;rintf("Command stored successfully.\n");
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ gcc -Wall topic1_history.c -o topic1
./topic1
Command History Demo
Type commands, :prev, :next, or :quit
> hellp
Command stored successfully.
> ls
Command stored successfully.
> :prev
Previous: ls
> :quit
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ cat > topic2_memory.c <<'EOF'
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};
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};

int main(void) {
    int capacity = 2, count = 0;
    int *arr = malloc(capacity * sizeof(int));

    if (arr == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    printf("Enter 5 integers:\n");

    while (count < 5) {
        int value;

        if (scanf("%d", &value) != 1) {
            printf("Invalid input\n");
            free(arr);
            return 1;
        }

        if (count == capacity) {
            capacity *= 2;
            int *temp = realloc(arr, capacity * sizeof(int));

            if (temp == NULL) {
                free(arr);
                printf("Resizing failed\n");
                return 1;
            }

            arr = temp;
EOF return 0;nMemory released successfully!\n");;Node));acity);
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ gcc -Wall -Wextra -std=c11 topic2_memory.c -o topic
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ ./topic2
Enter 5 integers:
1
2
3
Array resized to capacity 4
4
5
Array resized to capacity 8

Dynamic array: 1 2 3 4 5
Linked list: 1 2 3 4 5
Memory released successfully!
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ cat > topic3_tokenizer.c <<'EOF'
#include <stdio.h>
#include <string.h>

int main(void) {
    char input[500];
    char *tokens[100];
    int count = 0;

    printf("Enter a sentence or command:\n");

    if (fgets(input, sizeof(input), stdin) == NULL)
        return 1;

    input[strcspn(input, "\n")] = '\0';
```

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krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ gcc -Wall -Wextra -std=c11 topic3_tokenizer.c -o topic3
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ ./topic3
Enter a sentence or command:
copy   file.txt, to; folder

Total tokens: 4
Token 1: copy
Token 2: file.txt
Token 3: to
Token 4: folder
Token stream validated successfully.
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ cat > topic4_parser.c <<'EOF'
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <ctype.h>

int main(void) {
    char input[300];
    char *tokens[20];
    int count = 0;

    printf("Enter a command (print WORD / add NUM NUM): ");

    if (fgets(input, sizeof(input), stdin) == NULL)
        return 1;

    char *p = input;

    while (isspace((unsigned char)*p))
        p++;

    if (*p == '\0') {
        printf("Syntax Error: Empty command!\n");
        return 1;
    }

    char *token = strtok(p, " \t\r\n");

    while (token != NULL && count < 20) {
        tokens[count++] = token;
        token = strtok(NULL, " \t\r\n");
    }

    printf("\nParse Tree:\n");
    printf("Command\n");

    for (int i = 0; i < count; i++)
        printf("  |-- Token %d: %s\n", i + 1, tokens[i]);

    if (strcmp(tokens[0], "print") == 0) {
EOF return 0;n 1;yntax Error: Unknown command\n");e integers\n");
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ gcc -Wall -Wextra -std=c11 topic4_parser.c -o topic4
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ ./topic4
Enter a command (print WORD / add NUM NUM): print hello

Parse Tree:
Command
  |-- Token 1: print
  |-- Token 2: hello
Execution Result: hello
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ add twelve 30
Command 'add' not found, did you mean:
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    printf("\nParse Tree:\n");
    printf("Command\n");

    for (int i = 0; i < count; i++)
        printf("  |-- Token %d: %s\n", i + 1, tokens[i]);

    if (strcmp(tokens[0], "print") == 0) {
EOF return 0;n 1;yntax Error: Unknown command\n");e integers\n");
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ gcc -Wall -Wextra -std=c11 topic4_parser.c -o topic4
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ ./topic4
Enter a command (print WORD / add NUM NUM): print hello

Parse Tree:
Command
  |-- Token 1: print
  |-- Token 2: hello
Execution Result: hello
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ add twelve 30
Command 'add' not found, did you mean:
  command 'hadd' from snap root-framework (v6-36-04)
  command 'and' from deb and (1.2.2-4.2)
  command 'dd' from deb coreutils (9.4-3ubuntu6.2)
  command 'adb' from deb adb (1:33.0.3-2)
  command 'adb' from deb google-android-platform-tools-installer (34.0.4+1695033340)
  command 'axd' from deb afnix (3.8.0-1)
  command 'adr' from deb adr-tools (3.0.0-2)
  command 'aid' from deb id-utils (4.6.28-20200521ss15dab)
  command 'ldd' from deb libc-bin (2.39-0ubuntu8.7)
  command 'ad' from deb netatalk (3.1.15~ds-1ubuntu1)
  command 'atd' from deb at (3.2.5-1ubuntu1)
  command 'tdd' from deb devtodo (0.1.20+git20200830.0ad52b0-3)
  command 'ddd' from deb ddd (1:3.3.12-5.4)
  command 'pdd' from deb pdd (1.7-1)
See 'snap info <snapname>' for additional versions.
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ ls
topic1            topic2           topic3              topic4
topic1_history.c  topic2_memory.c  topic3_tokenizer.c  topic4_parser.c
krishnasree08@krishnasree08:~/ossp_ubuntu_topics$ lsls
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Result: Successfully implemented and executed all four C programs in Ubuntu, demonstrating command history, dynamic memory allocation, tokenization, and parsing with syntax validation.